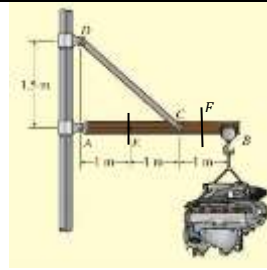


Example 1

The 500 Kg engine is suspended from the crane boom as shown. Determine the resultant internal loadings acting on the cross section of the boom at points E and F.

$$\tan \theta = \frac{1.5}{2} \gg \gg \theta = 37^\circ$$

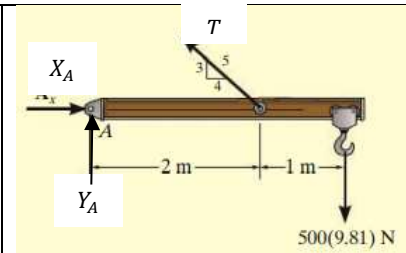


Consider the free body diagram of the boom AB.

$$\Sigma M_A = 0 \gg \gg T \sin \theta \times 2 - 4905 \times 3 = 0 \gg \gg T = 12225.5 \text{ N}$$

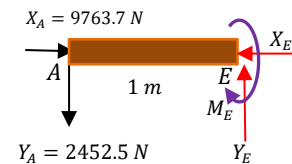
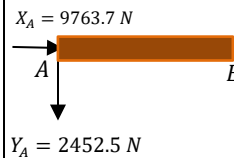
$$\Sigma F_x = 0 \gg \gg X_A - T \cos \theta = 0 \gg \gg X_A = 9763.7 \text{ N}$$

$$\Sigma F_y = 0 \gg \gg Y_A + T \sin \theta - 4905 = 0 \gg \gg Y_A = -2452.5 \text{ N}$$



To see the internal loading at E, let us cut the boom at E and consider the free body diagram of the left part.

For equilibrium, at E, there should be forces equal and opposite to X_A and Y_A



Now, the forces are balanced, but the two equal parallel vertical forces at A and E produces a couple in the ccw direction. Therefore, the forces in the cross-section at E should also produce a clockwise couple for the equilibrium.

$$\Sigma F_x = 0 \gg \gg X_A - X_E = 0 \gg \gg X_E = 9763.7 \text{ N}$$

$$\Sigma F_y = 0 \gg \gg Y_E - Y_A = 0 \gg \gg Y_E = 2452.5 \text{ N}$$

Algebraic sum of moments of all forces about the perpendicular axis passing through E

$$\Sigma \text{Moments}_E = 0 \gg \gg Y_A \times 1 - M_E = 0 \gg \gg M_E = 2452.5 \text{ Nm}$$

Similarly, consider the free body diagram of portion of boom AF

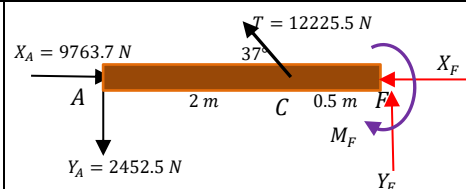
$$\Sigma F_x = 0 \gg X_A - X_F - T \cos 37 = 0 \gg \gg X_F = 0$$

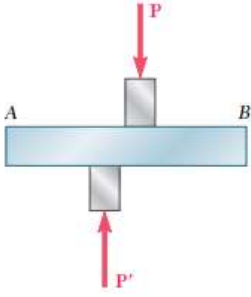
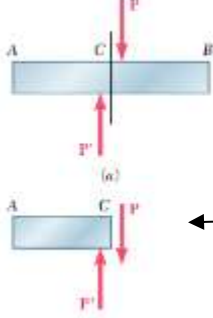
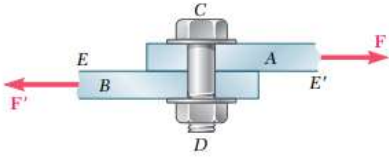
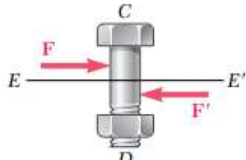
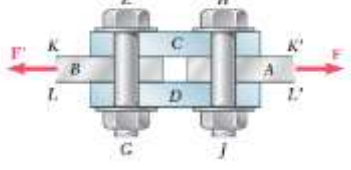
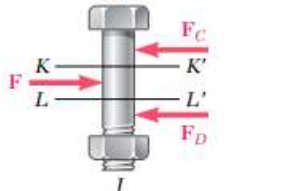
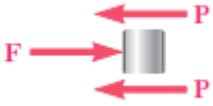
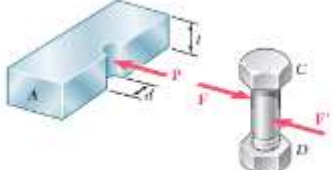
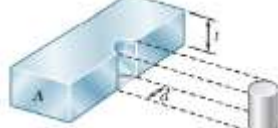
$$\Sigma F_y = 0 \gg \gg Y_F - Y_A + T \sin 37 = 0 \gg \gg Y_F = -4905 \text{ N}$$

Algebraic sum of moments of all forces about the perpendicular axis passing through F

$$\Sigma \text{Moments}_F = 0 \gg \gg$$

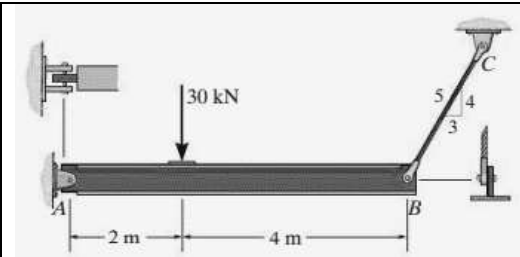
$$Y_A \times 2.5 \text{ m} - T \sin 37 \times 0.5 \text{ m} - M_F = 0 \gg \gg M_F = 2452.5 \text{ Nm}$$



		$\text{shear stress} = \frac{P}{A} = \frac{F}{A}$
	 <i>single shear</i>	
	 <i>double shear</i>	 $\text{shear stress} = \frac{F}{2A} = \frac{P}{A}$
$\text{Bearing stress} = \frac{P}{td}$		

Example 2

Determine the average shear stress in the 20 mm diameter pin at A and the 30 mm diameter pin at B that support the beam in the figure.



Let us find out the forces acting on the pins by considering the free body diagram of the beam.

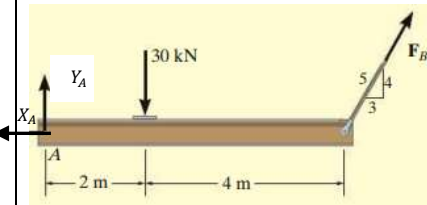
$$\Sigma M_A = 0 \gg F_B \sin \theta \times 6\text{ m} - 30\text{ kN} \times 2\text{ m} = 0 \gg F_B = 12.5\text{ kN}$$

$$\Sigma F_x = 0 \gg -X_A + F_B \cos \theta = 0 \gg X_A = 7.5\text{ kN}$$

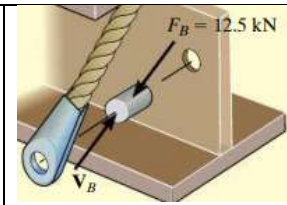
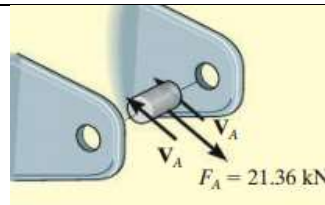
$$\Sigma F_y = 0 \gg Y_A + F_B \sin \theta - 30\text{ kN} = 0 \gg Y_A = 20\text{ kN}$$

The resultant force exerted by pin A on the beam,

$$F_A = \sqrt{7.5^2 + 20^2} = 21.36\text{ kN}$$



The same force F_A acts on the pin in the opposite direction. Now, we can see that the pin at A is in double shear and the pin at B is in single shear.

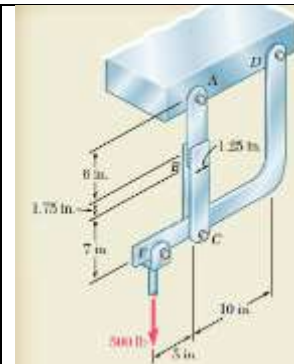


The average shear stress on pin A , $\tau_A = \frac{F_A}{2 \times \frac{\pi d^2}{4}} = \frac{21.36\text{ kN}}{2 \times \frac{\pi (0.02\text{ m})^2}{4}} = 34 \times 10^6\text{ N/m}^2$

The average shear stress on pin B , $\tau_B = \frac{F_B}{\frac{\pi d^2}{4}} = \frac{12.5\text{ kN}}{\frac{\pi (0.03\text{ m})^2}{4}} = 17.7 \times 10^6\text{ N/m}^2$

Example 3

In the hanger shown, the upper portion of the link ABC is $3/8$ in. thick and the lower portions are each $1/4$ in. thick. Epoxy resin is used to bond the upper and lower portions together at B . The pin at A is of $3/8$ in. diameter while a $1/4$ in. diameter pin is used at C . Determine the shearing stress in pin A and C , the largest normal stress in the link ABC , the average shear stress on the bonded surfaces at B , and the bearing stress in the link at C .



$$1\text{ in.} = 25.4\text{ mm}, \quad 1\text{ lb} = 4.45\text{ N}$$

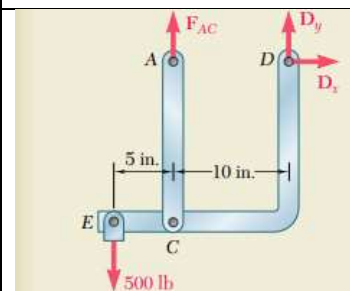
Consider the free body diagram of the hanger

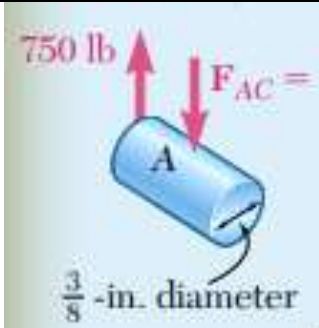
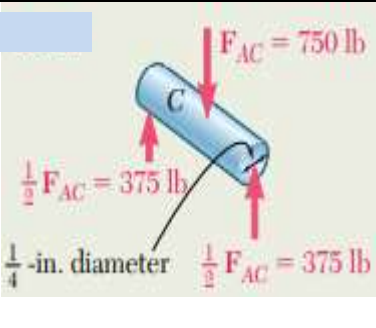
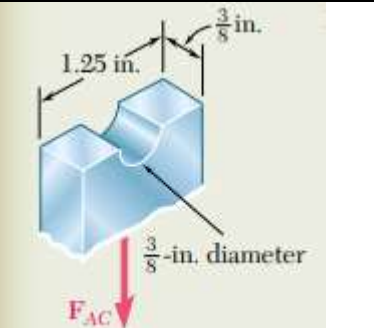
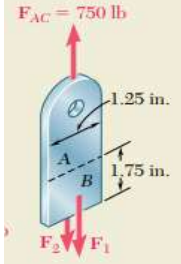
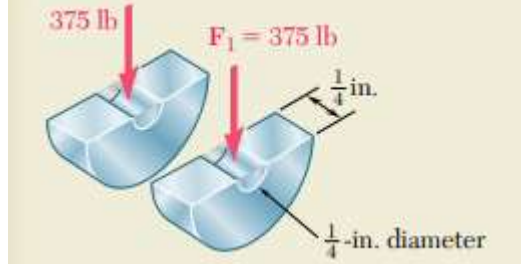
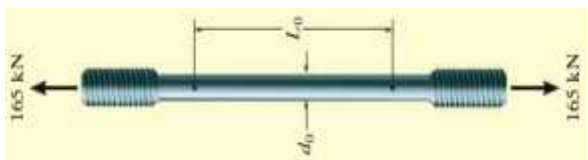
$$\Sigma M_D = 0 \gg -F_{AC} \times 10\text{ in} + 500\text{ lb} \times 15\text{ in} = 0 \gg$$

$$F_{AC} = 750\text{ lb} = 3337.5\text{ N}$$

$$\Sigma F_x = 0 \gg D_x = 0$$

$$\Sigma F_y = 0 \gg D_y + F_{AC} - 500\text{ lb} = 0 \gg D_y = -250\text{ lb} = -1112.5\text{ N}$$



		
The pin at A is in single shear. The average shear stress on the pin A, $\tau_A = \frac{750 \text{ lb}}{\frac{\pi \times (0.375 \text{ in})^2}{4}} = 6790 \text{ psi} = 46.84 \text{ MPa}$	The pin at C is in double shear. The average shear stress on the pin C, $\tau_c = \frac{750 \text{ lb}}{2 \times \frac{\pi \times (0.25 \text{ in})^2}{4}} = 7640 \text{ psi} = 52.69 \text{ MPa}$	The normal stress in the link ABC is maximum where the cross-sectional area is minimum. $\sigma_A = \frac{750 \text{ lb}}{(1.25 - 0.375) \text{ in} \times 0.375 \text{ in}} = 2285 \text{ psi} = 15.77 \text{ MPa}$
		
The bonded surfaces are in double shear. $\tau_B = \frac{750 \text{ lb}}{2 \times (1.25 \text{ in} \times 1.75 \text{ in})} = 171.4 \text{ psi} = 1.18 \text{ MPa}$	The bearing stress in link at C $\sigma_{\text{bearing}} = \frac{750 \text{ lb}}{2 \times (0.25 \text{ in} \times 0.25 \text{ in})} = 6000 \text{ psi} = 41.39 \text{ MPa}$	
Example 4 An aluminium specimen shown in figure has a diameter 25mm and gauge length 250 mm. If a force of 165 kN elongates the gauge length 1.20mm, determine the modulus of elasticity. Also, determine the contraction in diameter. Take $G_{al} = 26 \text{ GPa}$ and $\sigma_{\text{yield}} = 440 \text{ MPa}$. Calculate the percentage elongation and percentage reduction in area.		
The average normal stress, $\sigma = \frac{P}{A} = \frac{165 \times 10^3 \text{ N}}{\left(\frac{\pi \times 0.025^2}{4}\right) \text{ m}^2} = 336.1 \times 10^6 \text{ Pa} = 336.1 \text{ MPa}$		
The normal (axial/longitudinal) strain, $\epsilon_{\text{long}} = \frac{\delta}{L} = \frac{1.2 \text{ mm}}{250 \text{ mm}} = 4.8 \times 10^{-3}$		

The stress developed (336.1 MPa) is less than the yield stress (440 MPa) of the material, it is in the linear elastic range.

Using Hooke's law, the modulus of elasticity, $E = \frac{\sigma}{\epsilon} = \frac{336.1 \times 10^6}{4.8 \times 10^{-3}} = 70 \times 10^9 \text{ Pa}$

The shear modulus of aluminium $G = 26 \text{ GPa}$. Using the relation, $G = \frac{E}{2(1+\nu)}$

$$1 + \nu = \frac{E}{2G} = \frac{70 \times 10^9}{2 \times 26 \times 10^9} = 1.346 \gggg \gg \nu = 0.346$$

$$\nu = -\frac{\epsilon_{lat}}{\epsilon_{long}} \gggg \gg \epsilon_{lat} = -0.346 \times 4.8 \times 10^{-3} = -1.662 \times 10^{-3}$$

Change in diameter = $\epsilon_{lat} \times d = -1.662 \times 10^{-3} \times 25 \text{ mm} = -0.0415 \text{ mm}$

The new length, $L' = L + \delta = 251.2 \text{ mm}$

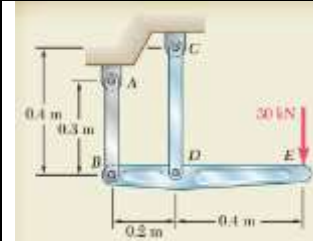
The new diameter, $d' = 25 - 0.0415 = 24.9585 \text{ mm}$

% increase in length = $\frac{251.2 - 250}{250} \times 100 = 0.48\%$

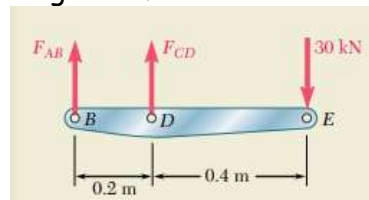
% reduction in area = $\frac{\frac{\pi}{4} \times 25^2 - \frac{\pi}{4} \times 24.9585^2}{\frac{\pi}{4} \times 25^2} \times 100 = 0.332\%$

Example 5

The rigid bar BDE is supported by two links AB and CD . Link AB is made of aluminium ($E = 70 \text{ GPa}$) and has a cross sectional area of 500 mm^2 ; link CD is made of steel ($E = 200 \text{ GPa}$) and has a cross sectional area of 600 mm^2 . For the 30 kN force shown, determine the deflection of B , D , and E



Consider the free-body diagram of bar BDE .



$$\sum M_B = 0$$

$$F_{CD} \times 0.2 \text{ m} - 30 \text{ kN} \times 0.6 \text{ m} = 0$$

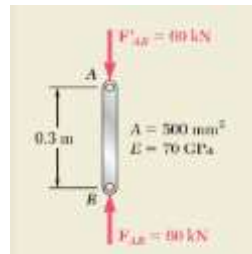
$$F_{CD} = +90 \text{ kN}$$

$$\sum F_Y = 0$$

$$F_{AB} + F_{CD} - 30 \text{ kN} = 0$$

$$F_{AB} = -60 \text{ kN}$$

Consider the bar AB



Change in length of AB
(displacement of B)

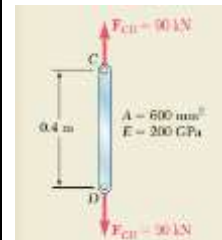
$$\delta_B = \frac{PL}{AE}$$

$$= -\frac{60 \times 10^3 \text{ N} \times 0.3 \text{ m}}{500 \times 10^{-6} \text{ m}^2 \times 70 \times 10^9 \text{ Pa}}$$

$$= -514 \times 10^{-6} \text{ m}$$

$$\delta_B = 0.514 \text{ mm} \uparrow$$

Consider the bar CD



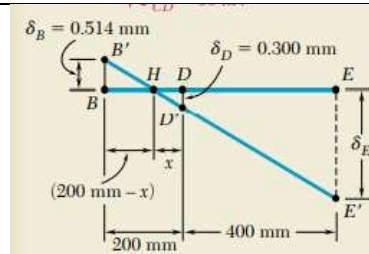
Change in length of CD
(displacement of D)

$$\delta_D = \frac{PL}{AE}$$

$$= \frac{90 \times 10^3 \text{ N} \times 0.4 \text{ m}}{600 \times 10^{-6} \text{ m}^2 \times 200 \times 10^9 \text{ Pa}}$$

$$= 300 \times 10^{-6} \text{ m}$$

$$\delta_D = 0.300 \text{ mm} \downarrow$$

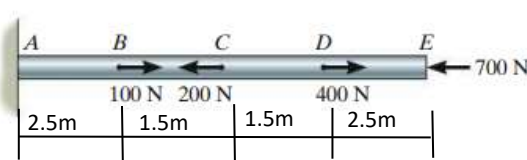


$$\frac{BB'}{DD'} = \frac{BH}{HD} \gggg \gg \frac{0.514 \text{ mm}}{0.300 \text{ mm}} = \frac{200 \text{ mm} - x}{x} \gggg \gg x = 73.7 \text{ mm}$$

$$\frac{BB'}{EE'} = \frac{BH}{HE} \gggg \gg \frac{0.514 \text{ mm}}{\delta_E} = \frac{200 \text{ mm} - 73.7 \text{ mm}}{400 \text{ mm} + 73.7 \text{ mm}} \gggg \gg \delta_E = 1.928 \text{ mm} \downarrow$$

Example 6

Determine the elongation of the steel bar ($E = 200 \text{ GPa}$, $A = 500 \text{ mm}^2$) which is loaded as shown.

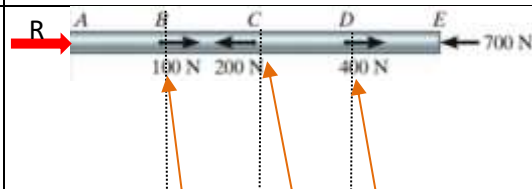


Consider the free-body diagram of the whole bar. For equilibrium,

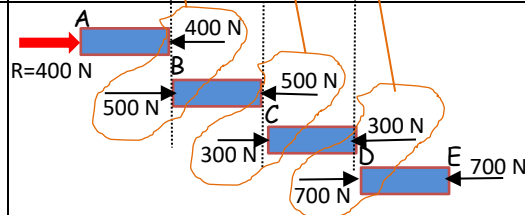
$$\Sigma F_x = 0$$

$$R + 100\text{N} - 200\text{N} + 400\text{N} - 700\text{N} = 0$$

$$R = 400\text{N}$$



Now, consider the equilibrium of the sections of the bar wherever there is a change in load or cross-sectional area.



Part AB: On the left side $R = (400 \text{ N} \rightarrow)$. Therefore, on the right side, there should be a force of 400 N in the opposite direction ($\leftarrow$) for equilibrium.

Part BC: Left side the force should be $(500 \text{ N} \rightarrow)$ so that when we combine AB and BC, there will be a net force of $(100 \text{ N} \rightarrow)$ at B. For equilibrium, the force on the right side should be $(500 \text{ N} \leftarrow)$

Part CD: Left side the force should be $(300 \text{ N} \rightarrow)$ so that when we combine BC and CD, there will be a net force of $(200 \text{ N} \leftarrow)$ at C. For equilibrium, the force on the right side should be $(300 \text{ N} \leftarrow)$

Part DE: Left side the force should be $(700 \text{ N} \rightarrow)$ so that when we combine CD and DE, there will be a net force of $(400 \text{ N} \rightarrow)$ at D. The force on the right side is $(700 \text{ N} \leftarrow)$.

Each segment will have an axial compressive deformation, $\delta = \frac{PL}{AE}$

$$\delta_{AB} = \frac{400\text{N} \times 2.5\text{m}}{500 \times 10^{-6} \text{m}^2 \times 200 \times 10^9 \text{Pa}} = 10 \times 10^{-6} \text{m} \quad \ll \gg \quad \delta_{BC} = \frac{500\text{N} \times 1.5\text{m}}{500 \times 10^{-6} \text{m}^2 \times 200 \times 10^9 \text{Pa}} = 7.5 \times 10^{-6} \text{m}$$

$$\delta_{CD} = \frac{300\text{N} \times 1.5\text{m}}{500 \times 10^{-6} \text{m}^2 \times 200 \times 10^9 \text{Pa}} = 4.5 \times 10^{-6} \text{m} \quad \ll \gg \quad \delta_{DE} = \frac{700\text{N} \times 2.5\text{m}}{500 \times 10^{-6} \text{m}^2 \times 200 \times 10^9 \text{Pa}} = 17.5 \times 10^{-6} \text{m}$$

Since we are considering deformations in the linear elastic range, we can apply the principle of superposition. Hence, the net deformation will be the algebraic sum of all the individual deformations.

$$\delta = -(\delta_{AB} + \delta_{BC} + \delta_{CD} + \delta_{DE}) = -39.5 \times 10^{-6} \text{m}$$

Example 7

Determine the strain energy stored in a prismatic bar suspended from one end due to its own weight, assuming linear elastic behavior.

The strain energy in the linear elastic region, (if load, cross sectional area, and elastic modulus are uniform)

$$U = \frac{1}{2} P \delta = \frac{1}{2} P \left(\frac{PL}{AE} \right) = \frac{P^2 L}{2AE} .$$

The strain energy density,

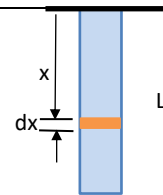
$$u = \frac{1}{2} \sigma \varepsilon = \frac{1}{2} \sigma \left(\frac{\sigma}{E} \right) = \frac{\sigma^2}{2E}$$

If the load, cross sectional area, and elastic modulus are varying with respect to its length,

$$U = \int_0^L \frac{P_x^2 dx}{2A_x E_x}$$

Let w be the weight density of the material in N/m^3 . Consider an elemental disc of the material having thickness dx at a distance x from the support. The force acting on this element is the weight of the bar below the element.

$$P_x = wA(L - x)$$

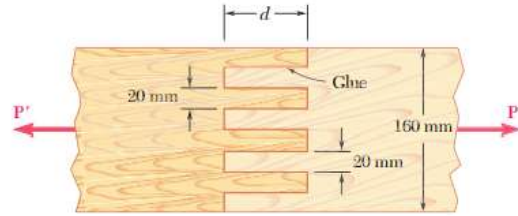


Since the cross sectional area and the elastic modulus are constant

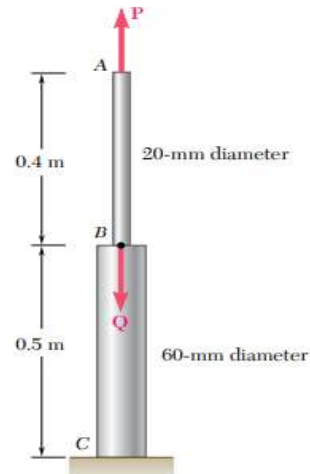
$$U = \int_0^L \frac{P_x^2 dx}{2A_x E_x} = \frac{1}{2AE} \int_0^L \{wA(L - x)\}^2 dx = \frac{w^2 A^2}{2AE} \left[L^2 x - 2L \frac{x^2}{2} + \frac{x^3}{3} \right]_0^L = \frac{w^2 L^3 A}{6E}$$

Exercise 1

Two wooden planks, each 22 mm thick and 160 mm wide, are joined by the glued mortise joint as shown. Knowing that the joint will fail when the average shearing stress in the glue reaches 820 kPa, determine the smallest allowable length d of the cuts if the joint is to withstand an axial load of magnitude $P = 7.6 \text{ kN}$

**Exercise 2**

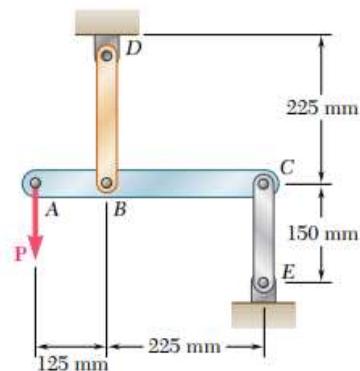
The rod ABC is made of an aluminium for which $E = 70 \text{ GPa}$. Knowing that $P = 6 \text{ kN}$ and $Q = 42 \text{ kN}$, determine the deflection of (i) point A and (ii) point B

**Exercise 3**

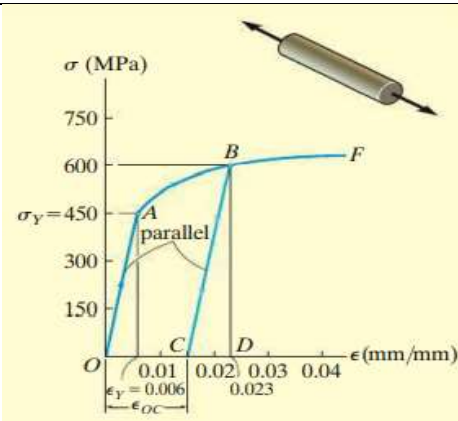
What should be the value of Q so that the deflection at A is zero? What will be the corresponding deflection of B?

Exercise 4

Link BD is made of brass ($E = 105 \text{ GPa}$) and has a cross-sectional area of 240 mm^2 . Link CE is made of aluminium ($E = 72 \text{ GPa}$) and has a cross-sectional area of 300 mm^2 . Knowing that they support a rigid member ABC, determine the maximum force P that can be applied vertically at point A if the deflection of A is not to exceed 0.35 mm .

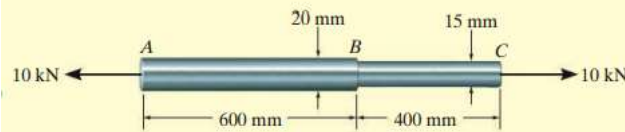
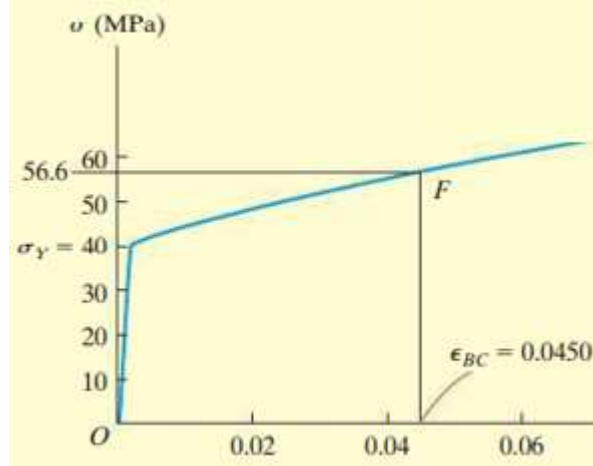
**Exercise 5**

The stress-strain diagram for an aluminium alloy that is used for making aircraft parts is shown in figure. If a specimen of the material is stress to 600 MPa , determine the permanent strain when the load is released. Also, find modulus of resilience both before and after the load application

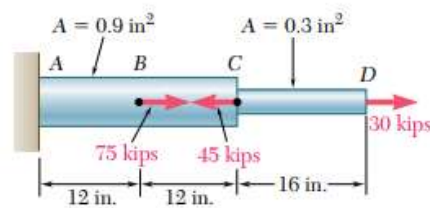


Exercise 6

Aluminium rod, shown in figure has a circular cross-section and is subjected to an axial load of 10 kN . If a portion of the stress strain diagram is shown, determine the elongation of the rod when the load is applied. Take $E = 70\text{ GPa}$

**Exercise 7**

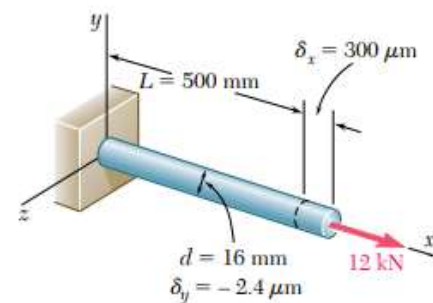
Determine the elongation of the bar if $E = 200\text{ GPa}$
($1\text{ kip} = 4.45\text{ kN}$, $1\text{ in} = 25.4\text{ mm}$)

**Exercise 8**

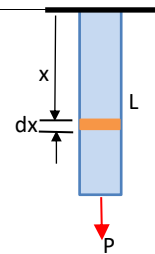
Determine the elongation of the conical bar due to its self-weight when suspended vertically if the material has a modulus of elasticity E and specific weight γ

**Exercise 9**

A 500 mm long, 16 mm long made of a homogeneous, isotropic material is observed to increase in length by $300\text{ }\mu\text{m}$, and to decrease in diameter by $2.4\text{ }\mu\text{m}$ when subjected to an axial load of 12 kN . Determine E and ν of the material.

**Exercise 10**

Determine the strain energy in a prismatic bar suspended from one end, if in addition to its own weight, it supports a load P at the lower end.



Reference:

1. *Mechanics of Materials*, 2/e, J. M. Gere & S. P. Timoshenko, CBS Publishers.
2. *Mechanics of Materials*, F.P. Beer & E. R. Johnston
3. *Engineering Mechanics of Solids*, 2/e, E. P. Popov, Pearson Education Asia.
4. *Mechanics of Materials*, 9/e, R. C. Hibbeler, Prentice Hall.